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Literature Review #4

Summary of Research Conducted by:

Charles Y. Shao, Rachna Sondhi, Paula S. van de Nes, and Todd Sacktor.
PKM ζ is both Necessary and Sufficient for Synaptic Clustering of PSD-95

Main Arguments Elicited by Shao et. al.

It has been established by prior research that the induction and maintenance of long term potentiation (LTP) and consequently, the maintenance of a stable memory trace are dependent on structural modifications of synapses via the modification of post-synaptic densities. The structural protein, PSD-95, has been implicated in several different aspects of this process including anchoring both neurotransmitter receptors and post-synaptic signaling molecules at the post synaptic density including AMPA receptors involved in LTP. However; before this study, it was unclear as to how the persistent structural reorganization of PSD-95 was mediated. In their research, the authors elucidated that the kinase PKM ζ is both necessary and sufficient for mediating the clustering of the structural scaffolding protein PSD-95.

Evidence for the role of PKM ζ in the Persistent Structural Reorganization of PSD-95

In order to prove PKM ζ 's role in the clustering of PSD-95 within the post synaptic density, the authors induced a chemical long term potentiation (cLTP) in harvested primary hippocampal neurons by subjecting these cultured cells to forskolin for 45 minutes. The hippocampal neurons were subsequently immune-stained with antibodies against PSD-95 and PKM ζ . The results of this test showed that the induction of cLTP caused the both the colocalization of PSD-95 and PKM ζ , and the clustering of PSD-95 at the post-synaptic density (Figure 1). The authors also elucidated that the application of a PKM ζ -specific inhibitor (ZIP) to the tissue prior to the immunostaining procedure prevented both the colocalization of PSD-95 with PKM ζ and the clustering of PSD-95 (Figure 1). The inhibition of PSD-95 clustering by another known PKM ζ -specific inhibitor, chelerythrine, proved that ZIP was not having some unforeseen effect like acting to inhibit PSD-95 clustering through a PKM ζ -independent pathway (Also figure 1). Upon subsequent statistical analysis, all of these results were proven to be significant. --- So basically in this part of the experiment, the authors proved that inducing cLTP causes the PSD-95 clusters and that PKM ζ was a necessary part of the pathway (when PKM ζ was inhibited, the clusters could not form, despite the induction of cLTP).

Next, the researchers set out to elucidate whether or not PKM ζ was *sufficient* to cause the clustering of PSD-95 and subsequent structural changes hallmark of long term potentiation. They accomplished this by transfecting hippocampal neurons with a Sindbis viral vector containing a GFP-PKM ζ transcript. They did this to see if the expression of PKM ζ could cause the PSD-95 clustering and subsequent AMPA receptor trafficking in the absence of cLTP. Their results were that the transfection of neurons with GFP- PKM ζ was indeed sufficient to cause these changes

(figure 2). Next, the researchers transfected the neurons to GFP alone, as well as GFP bound to an inactive PKM ζ analog to prove that their results were due to the action of PKM ζ and not their transfection procedure, which they found to be true (figure 2). After that, the researchers applied the ZIP (the PKM ζ inhibitor) to the neurons after the transfection procedure reversed the clustering and proved that the clustering was due to the actions of PKM ζ (figure 2). Finally, the researchers proved that the transfection of neurons with the GFP- PKM ζ caused an increase in spine size and resulted in the trafficking of AMPA-GluA2 receptors to the PSD (figure 2), which proved that PKM ζ is sufficient to cause the characteristic changes of LTP *en vitro* without the induction of cLTP.

Relation to *Neurobiology of Learning and Memory* by J.W. Rudy:

The chapters of *The Neurobiology of Learning and Memory* that we have read thus far have chronicled the changes to the post synaptic density caused by the induction of LTP; the stabilization of these changes following strong inducing stimuli; and the subsequent consolidation and maintenance of these modifications leading to the expression of a stable memory trace. Specifically, chapter 7 in *The Neurobiology of Learning and Memory* talks about the specific role that PKM ζ has in the maintenance of the changes in dendritic spines subsequent to consolidation of LTP. *NLM* talks about the role PKM ζ plays in the trafficking of GluA2 receptors to the PSD, and PKM ζ 's inhibition of the endocytic mechanisms that normally remove these receptors from the PSD. This research elucidates another role PKM ζ could play in the maintenance of LTP – through the mediation of PSD-95.

Implications for the Future of Learning and Memory:

Studies thus far have shown that PKM ζ plays a role in the maintenance of memory. Now the research being conducted not only by the authors of this paper but by others as well is attempting to elucidate the specific mechanistic action of PKM ζ and other proteins in facilitating this process. Thus if we discover the full mechanistic action as to how memory is maintained within our brains we can potentially use this knowledge for the development of potential therapeutics towards diseases like dementia where this process is known to break down. Furthermore, possible analogs of PKM ζ could serve as memory enhancing supplements in the future. Overall, further research is needed in this field and it will be exciting to see where this research takes us.